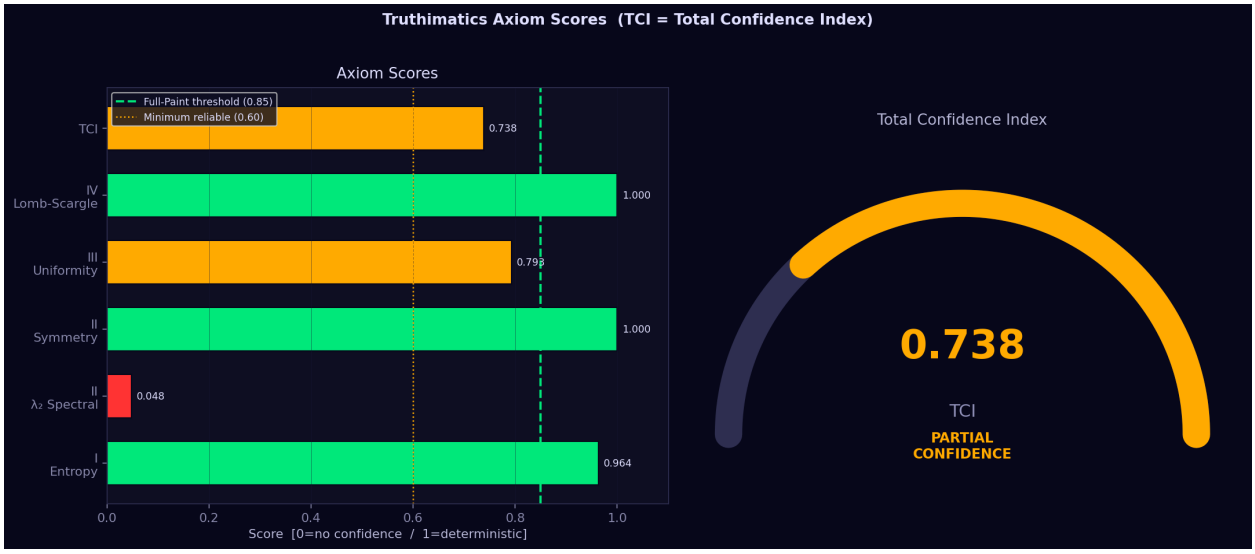
AXIOM-DINOSAUR v4.0

Truthimatics Full-Paint Deterministic Synthesis

Specimen ID	AXIOM_35F2A75C3C
Taxa	Velociraptor
Clade	Theropod
PBDB Records	19
Mass (regression)	39.9 kg (Campione & Evans 2012)
Mass (volumetric)	70.5 kg (Silhouette integral, ρ=1060.0 kg/m^3)
Body Volume	0.083 m^3
Validity Score	1.0000 / 1.0000
TCI	0.7382 (partial)
Axiom I Entropy	0.9636
Axiom II λ■	0.1937 (symmetry 100%)
Axiom III Uniformity	0.7931
Axiom IV Lomb FAP	1.000000 peak 0.000 Hz
Integrity	60.0% Measured / 40.0% Inferred
Elements	25
CoG (m)	[0.082, 0.154, 0.0]
Generated	2026-04-30 23:53 UTC

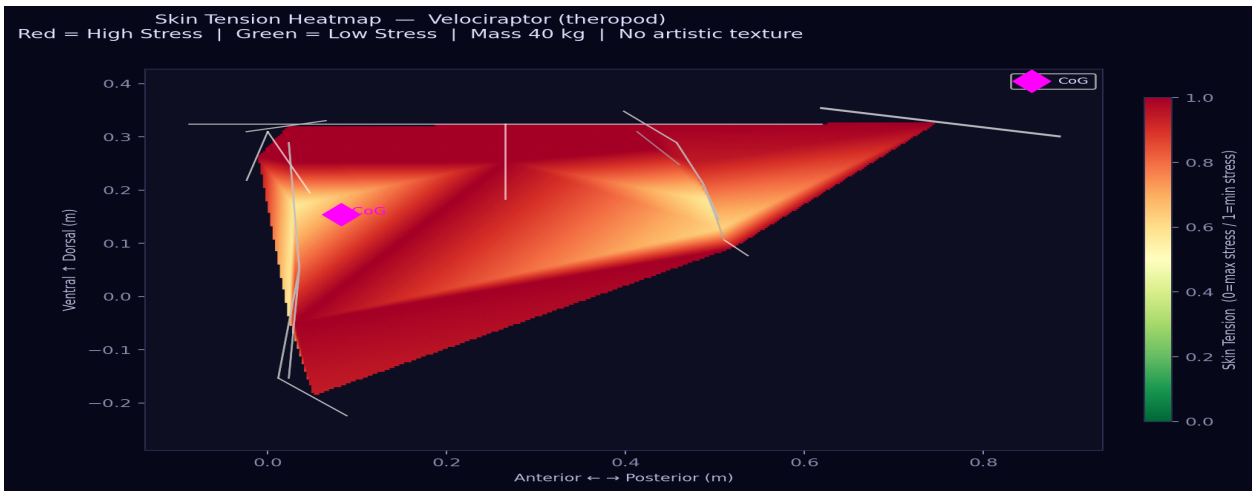
1 Truthimatics Confidence Analysis

TCI = weighted sum: Axiom I (entropy, w=0.25) + Axiom II (spectral, w=0.30) + Axiom III (Biewener uniformity, w=0.30) + Axiom IV (Lomb, w=0.15). TCI ≥ 0.85 authorises Full-Paint output.



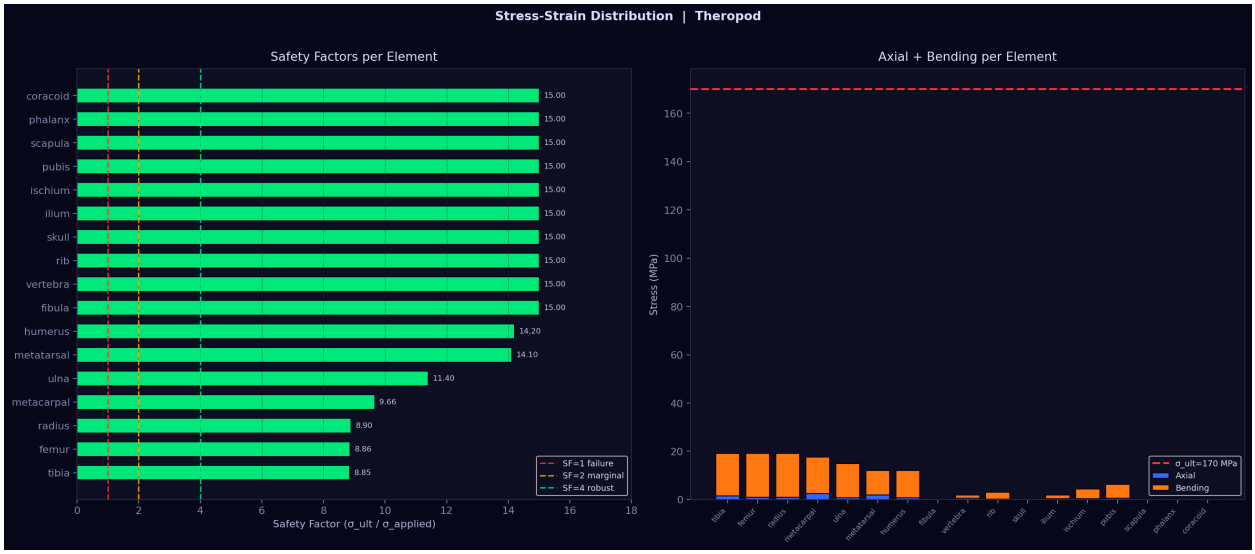
2 Skin Tension Heatmap

Stress field interpolated from per-bone safety factors onto the 2D body silhouette. Red=high stress (low SF), Green=low stress (high SF). No artistic texture applied.



3 Stress-Strain Distribution

Loading: biped (F=Mg). σ_ult=170 MPa (Reilly & Burstein 1975). SF≥4 robust.

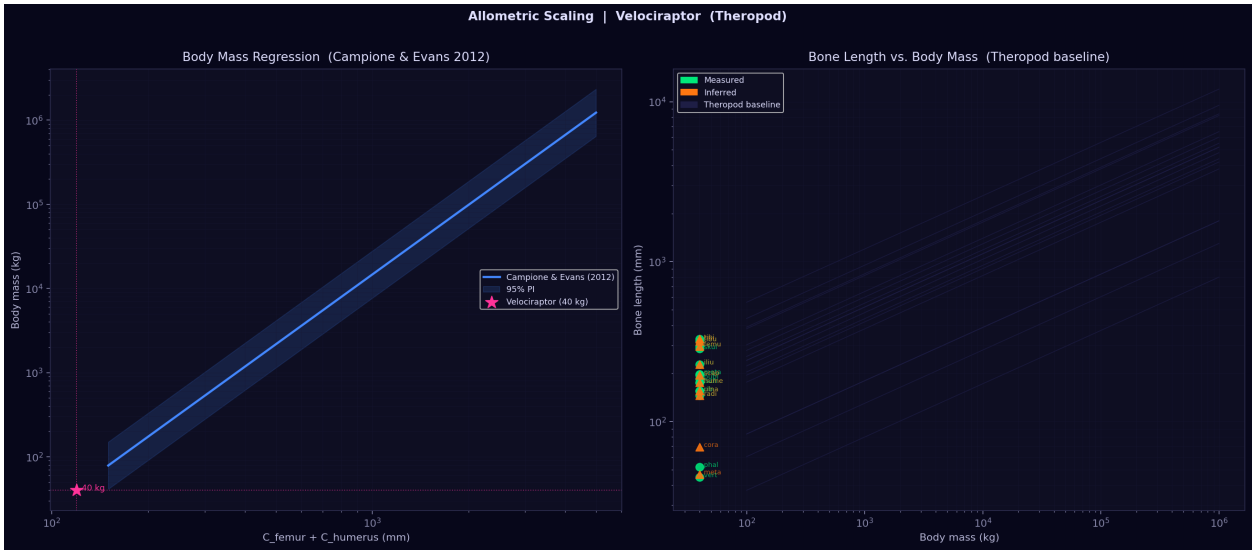


Per-Element Stress Summary

Element	σ_ax MPa	σ_bend MPa	σ_total MPa	SF	Status
coracoid	0.054	0.140	0.194	15.000	OK
femur	1.010	18.170	19.179	8.864	OK
fibula	0.000	0.000	0.000	15.000	OK
humerus	0.763	11.208	11.972	14.200	OK
ilium	0.277	1.615	1.892	15.000	OK
ischium	0.428	4.017	4.445	15.000	OK
metacarpal	2.502	15.091	17.593	9.663	OK
metatarsal	1.909	10.145	12.054	14.104	OK
phalanx	0.000	0.000	0.000	15.000	OK
pubis	0.528	5.825	6.353	15.000	OK
radius	0.844	18.263	19.107	8.897	OK
rib	0.083	2.964	3.047	15.000	OK
scapula	0.023	0.108	0.130	15.000	OK
skull	0.000	0.000	0.000	15.000	OK
tibia	1.508	17.705	19.213	8.848	OK
ulna	0.684	14.232	14.916	11.397	OK
vertebra	0.623	1.344	1.968	15.000	OK

4 Allometric Scaling Analysis

Campione & Evans (2012): log■■(M_g)=2.754×log■■(C_f+C_h)–1.097 (R^2=0.97; N=426 non-avian dinosaurs).



5 Physical Properties Analysis

Biomechanical performance estimates derived deterministically from skeletal geometry. Speed: Alexander (1976) locomotion equation. Bite force: Christiansen & Wroe (2007). All values represent physically constrained estimates, not artistic approximations.

Parameter	Value	Notes
Hip Height	0.821 m	Femur + tibia + metatarsal
Estimated Speed	9.0 km/h (2.5 m/s)	Gait: Trot | Fr=0.775
Stride Length	1.744 m	Alexander (1976) / Thulborn (1990)
Bite Force	200.2 N (20.4 kgf)	Calibrated theropod regression (Bates & Falkingham 2012)
Skull Length	285.0 mm	Mid-shaft proxy from skull feature vector
Femoral Robusticity	0.2343	Circumference / length (dimensionless)
Limb Strength Index	2.7e-05	(A_f + A_t) / M^0.67 (Christiansen 1998)
Est. Body Length	2.7 m	Allometric from femur length
Est. Neck Length	0.38 m	Allometric from femur length
Est. Tail Length	1.03 m	Allometric from femur length

6 Complete Bone Manifest

Element	L (mm)	D (mm)	Vol cm³	Mass g	Method	Conf	SF
coracoid	69.2	35.6	35.22	44.0	Constraint	0.40	15.00
femur	295.0	22.0	57.19	71.4	Measured	0.88	8.86
femur_right	295.0	22.0	57.19	71.4	Symmetry	0.84	8.86
fibula	318.0	7.0	4.41	4.6	Measured	0.88	15.00
fibula_right	318.0	7.0	4.41	4.6	Symmetry	0.84	15.00
humerus	175.0	16.0	17.94	22.4	Measured	0.88	14.20
humerus_right	175.0	16.0	17.94	22.4	Symmetry	0.84	14.20
ilium	228.0	42.0	161.10	201.1	Measured	0.88	15.00
ilium_right	228.0	42.0	161.10	201.1	Symmetry	0.84	15.00
ischium	175.0	20.0	28.04	35.0	Measured	0.88	15.00
metacarpal	46.8	8.6	0.51	0.4	Constraint	0.65	9.66
metatarsal	198.0	16.0	20.30	25.3	Measured	0.88	14.10
phalanx	52.0	10.0	0.63	0.5	Measured	0.88	15.00
pubis	185.0	18.0	24.01	30.0	Measured	0.88	15.00
radius	145.0	9.0	4.70	5.9	Measured	0.88	8.90
radius_right	145.0	9.0	4.70	5.9	Symmetry	0.84	8.90
rib	155.0	7.0	3.04	3.8	Measured	0.88	15.00
scapula	195.0	55.0	236.28	294.9	Measured	0.88	15.00
scapula_right	195.0	55.0	236.28	294.9	Symmetry	0.84	15.00
skull	285.0	110.0	1381.31	1724.3	Measured	0.88	15.00
tibia	328.0	18.0	42.57	53.1	Measured	0.88	8.85
tibia_right	328.0	18.0	42.57	53.1	Symmetry	0.84	8.85
ulna	155.0	10.0	6.21	7.8	Measured	0.88	11.40
ulna_right	155.0	10.0	6.21	7.8	Symmetry	0.84	11.40
vertebra	45.0	28.0	14.13	17.6	Measured	0.88	15.00